

Teaching Note

Capital Budgeting Microsimulation

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1. Simulation Overview

Simulation Brief

In this single-player simulation, participants manage capital budgeting for an Indian insurance software company with INR 68 million to allocate across 14 potential investment projects spanning four strategic categories (Market Expansion, Operational Efficiency, Process Improvement, and Product Expansion). Each project has distinct risk profiles, cash flow patterns, investment requirements, and time horizons.

Learning Objectives

- Apply quantitative capital budgeting techniques (NPV, IRR, PI, payback)
- Develop risk-adjusted evaluation methods for projects with varying risk profiles
- Construct an optimized project portfolio within budget constraints
- Analyse macroeconomic influence on capital allocation decisions
- Formulate a coherent capital budgeting strategy

Time Required: 15-20 mins prep, 45-60 mins simulation, 45-60 mins debrief

2. Essential Preparation Elements

Key Information (Instructor-Only):

- ✓ The macro-economic indicators are included to provide a broader outlook on the simulation; however, they do not directly impact the discount rate percentage decision under the Project Portfolio Decision Tab.
- ✓ These Macroeconomic indicators become relevant only during the project selection phase in the second decision tab, where participants evaluate projects based on external economic conditions and the financial metrics.
- ✓ The Weighted Average Cost of Capital (WACC) is provided under Company Indicators in the Market section. Participants can use this as a benchmark (hurdle rate) to guide their discount rate decisions, but this should be self-discovered through analysis rather than explicitly communicated.
- ✓ The discount rate percentage set in the Project Portfolio Decision area dynamically influences financial metrics such as NPV, Benefit-Cost Ratio, Profitability Index, and

EAC in the Decide tab. Participants should leverage these insights to refine their project selection strategy.

- ✓ The final report KPIs are calculated based on the actual discount rate percentage determined by the system, reflecting the financial impact of the participant's decisions.

Indicators (Available under the Market section of the Platform)

Macroeconomic Indicators	
Expected Repo Rate by RBI	5.5%
Inflation Rate	4.0%
GDP Growth Rate	7.5%
Unemployment Rate	6.0%
Exchange Rate (USD to INR)	75.5
Company-Specific Indicators	
Cost of Equity	13.0%
Cost of Debt	9.0%
Debt-to-Equity Ratio	40.0%
Tax Rate	25.0%
Weighted Average Cost of Capital	10.5%
Budget (INR Million)	68

Calculations

- Discount rate = Hurdle rate (WACC)*Multiplier of the project based on risk; multiplier values are provided in instructor account >> click on card >> case management >> project portfolio
- Base WACC calculation: 10.5 % (derived from company's capital structure)
- Benefit-cost ratio = NPV/Initial Investment
- NPV = Present value of future cash flow - Initial Investment in present
- EAC = Initial Investment/Annuity Factor, where Annuity factor = $(1 - 1/(1+\text{Discount rate})^{\text{period}})/\text{Discount rate}$.

Calculation on Base WACC calculation is as follows:

$$\text{Formula for WACC} = (E/V * Re) + ((D/V * Rd * (1-Tc))$$

Where:

- E/V = Equity proportion (from Debt-to-Equity ratio)
- D/ V = Debt proportion (from Debt-to-Equity ratio)

- R_e = **Cost of equity** (return required by equity investors)
- R_d = **Cost of debt** (interest rate on debt)
- T_c = **Tax rate** (since interest on debt is tax-deductible)

Given **Debt-to-Equity ratio = 40%**, we assume:

- Equity = 100
- Debt = 40
- **Total Capital (V)** = Equity + Debt = 100 + 40 = 140
- Therefore:
 - $E/V = 100/140 = 0.714$ (Equity Proportion)
 - $D/V = 40/140 = 0.286$ (Debt Proportion)
 - Apply the WACC Formula
 - $WACC = (E/V * R_e) + (D/V * R_d * (1 - T_c))$
 - $WACC = (0.714 \times 13\%) + (0.286 \times 9\% \times (1 - 25\%)) = 11.21\%$

By adjusting key financial factors, the base **WACC is considered to be 10.5%** in the simulation. Modifications in **market risk assumptions**, while slight deviations in the **debt-to-equity ratio** impacted capital weighting. Additionally, the influence of **macroeconomic conditions** such as the **repo rate (5.5%)** and **inflation (4.0%)**, along with internal risk assessments, resulted in a more **conservative** WACC estimate, ensuring alignment with market conditions and risk optimization strategies.

- Risk adjustment multipliers:
 - Low risk: 0.9 (discount rate), 1.0 (cash flow)
 - Low-moderate: 1.02 (discount rate), 0.97 (cash flow)
 - Moderate: 1.05 (discount rate), 0.95 (cash flow)
 - Moderate-high: 1.07 (discount rate), 0.93 (cash flow)
 - High risk: 1.09 (discount rate), 0.91 (cash flow)

Pedagogical Strategy:

- Students do not see exact risk multipliers, creating realistic information asymmetry
- This uncertainty requires estimation and analysis, mirroring real-world decisions
- During debrief, revealing these parameters creates powerful learning moments.

3. Facilitation Roadmap

Pre-Simulation (10 mins):

- Frame challenge: "As CFO, allocate limited capital across competing projects"
- Review key concepts: NPV, IRR, Payback, risk adjustment
- Explain evaluation process and interface
- Note: "Like real executives, you'll decide with incomplete information"

During Simulation (45-60 mins):

- 15 mins: Reminder about risk adjustment methodologies
- 30 mins: Suggestion to review budget utilization
- 45 mins: Prompt to consider portfolio balance.

Guiding Without Revealing: When students ask about precise calculation methods, redirect with questions:

- "What range of discount rate adjustments would be reasonable?"
- "How might you test sensitivity to different assumptions?"
- "What signals in project descriptions help gauge relative risk?"

Debrief Structure (45-60 mins):

1. Opening Discussion (10 mins):

- "What approach did you take to project evaluation?"
- "How did you balance risk and return?"

2. Results Analysis (15 mins):

- Display aggregate class results
- Highlight exemplar portfolios (highest NPV, best risk balance, etc.)
- Reveal actual risk parameters and their impact

3. Decision Frameworks (20 mins):

- WACC calculation and risk adjustment
- Project ranking methods (PI, IRR, etc.)
- Portfolio construction approaches

4. Synthesis and Application (15 mins):

- Connect to real-world capital budgeting challenges
- Discuss information asymmetry and estimation
- Explore how approach would change under different conditions.

4. Critical Decision Points & Best Practices

Discount Rate Determination:

Here are the risk-adjusted WACC rates based on the simulation's **10.5% WACC**:

- **Low risk: 9.45%** ($10.5\% \times 0.9$)
- **Low to moderate risk: 10.71%** ($10.5\% \times 1.02$)
- **Moderate risk: 11.03%** ($10.5\% \times 1.05$)
- **Moderate to high risk: 11.24%** ($10.5\% \times 1.07$)
- **High risk: 11.45%** ($10.5\% \times 1.09$)

Cash Flow Adjustment:

Risk-adjusted cash flow = Expected cash flow $\times$ Cash flow multiplier

Example (high-risk): $22.5\text{M} \times 0.91 = 20.48\text{M INR}$

Portfolio Optimization:

- Calculate risk-adjusted NPV for all projects
- Compute Profitability Index (PI) for each project
- Rank projects by PI in descending order
- Select projects in rank order until budget is exhausted
- Consider substitutions to maximize value within budget constraint

5. Sample Optimal Solution

It's important to recognize that there are many possible combinations that can help participants achieve the desired outcomes. The solution provided below is merely one example for faculty reference and is not meant for direct communication with students.

Project Portfolio Discount Rate	
Global Outreach Initiative	11.03%
Regional Domination Campaign	9.45%
Digital Transformation Drive	11.45%
Emerging Markets Expansion	11.45%
Efficiency Overhaul Program	11.03%
Supply Chain Optimization	9.45%
Employee Training Enhancement	10.71%
Automated Claims Processing	11.24%
Customer Onboarding Redesign	10.71%
Regulatory Compliance Enhancement	11.45%
Data Security Strengthening	11.45%

Innovative Insurance Products	11.03%
Partnership for New Market Entry	11.24%
Customizable Policy Options	10.71%
Project Portfolio Selection	
Global Outreach Initiative	Not Selected
Regional Domination Campaign	Selected
Digital Transformation Drive	Not Selected
Emerging Markets Expansion	Not Selected
Efficiency Overhaul Program	Not Selected
Supply Chain Optimization	Selected
Employee Training Enhancement	Not Selected
Automated Claims Processing	Selected
Customer Onboarding Redesign	Selected
Regulatory Compliance Enhancement	Not Selected
Data Security Strengthening	Not Selected
Innovative Insurance Products	Selected
Partnership for New Market Entry	Not Selected
Customizable Policy Options	Not Selected

Portfolio Analysis:

- Budget utilization: 98.16% - 66.8 million
- Category allocation: Market Expansion: 16.18%, Operational Efficiency: 13.24%, Process Improvement: 41.18%, Product Expansion: 27.94%.
- Risk profile: Low (29.85%), Low-Moderate (13.43%), Moderate (28.36%), Moderate to High (28.36%)

6. Common Pitfalls and Interventions

Student Pitfall	Recognition Signs	Intervention
Ignoring time value of money	Selecting based on absolute returns	Mini-lecture on NPV with timeline visualization
Risk adjustment errors	Inconsistent application of risk factors	Demonstrate process with sample project
Budget maximization issues	Large unallocated budget	Guide through incremental optimization
Category concentration	>60% allocation to single project type	Discuss risks of over-concentration
Overemphasis on financial metrics	Neglect of strategic considerations	Introduce strategic alignment matrix

Teaching Insight: Frame information asymmetry errors as realistic challenges executives face, using them to discuss techniques for decision-making under uncertainty.

7. Professor Tips

- ✓ **Avoid** asking "Who had the highest NPV?" at debrief start. Instead ask about balancing multiple objectives.
- ✓ When students comment on incomplete information, use it as a teaching moment about real executive decision-making.
- ✓ Reveal actual risk parameters gradually during debrief for maximum impact.
- ✓ For executives, periodically ask, "How does this compare to capital budgeting in your organization?"
- ✓ Watch for students finishing very quickly – they may have missed key analytical steps.
- ✓ This simulation's deliberate information asymmetry creates a realistic decision environment that challenges students to estimate parameters, test assumptions, and consider multiple scenarios – developing critical skills that transfer directly to real-world business contexts.

Discussion Starters

Macroeconomic & Company-Specific Indicators

1. Should macroeconomic conditions (e.g., inflation, GDP growth) have a stronger influence on project selection than company-specific indicators?
2. What impact does inflation have on the real value of future cash flows, and how should companies account for it?
3. How does GDP growth rate affect the potential success of market expansion projects?
4. Why is the unemployment rate relevant when selecting projects, particularly for process improvement or operational efficiency?
5. How can exchange rate fluctuations impact the profitability of international projects?
6. What role does the cost of equity play in setting a project's discount rate?
7. How does the cost of debt influence project financing decisions?
8. Why is the debt-to-equity ratio important in assessing a company's financial flexibility for new projects?
9. Why is the Weighted Average Cost of Capital (WACC) used as a benchmark for selecting projects?

10. If the exchange rate (USD to INR) depreciates further, how would that impact projects reliant on imported technology or resources?
11. How does a company's capital structure (debt-to-equity ratio) influence its ability to take on high-risk, high-return projects?

Project Selection & Financial Metrics

12. What factors should influence the selection of the discount rate for each project?
13. How does the risk level of a project impact its expected return?
14. How does the choice of discount rate impact the financial feasibility of a project?
15. What are the trade-offs between selecting high-NPV projects versus lower-risk projects?
16. Is a project with a shorter payback period always better than one with a longer payback period? Why or why not?
17. In a high-inflation environment, should companies prioritize projects with faster payback periods? Why or why not?
18. Why is NPV considered more reliable than IRR for decision-making?
19. What are some limitations of using financial metrics alone for project evaluation?
20. If two projects have similar NPVs but different risk levels, how should the company decide?

Strategic Decision-Making & Capital Allocation

21. How should companies prioritize between market expansion and operational efficiency projects?
22. If a company has a limited budget, should it focus on fewer high-return projects or diversify across multiple smaller projects?
23. What external factors (such as regulatory changes or global economic shifts) could significantly impact the success of the selected projects?
24. Why is it important to consider opportunity cost when selecting projects?
25. What are the benefits of diversifying investments across different types of projects?